

Lesson 9.3: Non-zero-sum games: cooperation

Game theory

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Collective vs individual interests

Example (Prisoners' dilemma)

Two armed robbers are making their get-away in a stolen car when the police pulls them over. The police doesn't have enough evidence to charge them with armed robbery but arrests them for car theft. The police interrogates the two suspects separately. Initially, both suspects remain silent (S), so the police offers each a reduced sentence if they 'talk/testify' (T) against their accomplice.

sentence	years
armed robbery	5
armed robbery, reduced	3
car theft	2
accessory	1

Individual player's perspective

you	accomplice	your sentence	years
remain silent	remain silent	theft	2
remain silent	testify	armed robbery	5
testify	remain silent	accessory	1
testify	testify	armed robbery, reduced	3

You (row player):

	t	s
T	3	1
S	5	2

Accomplice (column player):

	t	s
T	3	5
S	1	2

Individual vs collective preference

	t	s
T	3, 3	1, 5
S	5, 1	2, 2

- ▶ T dominates S
- ▶ t dominates s
- ▶ (T, t) equilibrium; value 3 for each player
- ▶ (T, t) is *Pareto dominated* by (S, s); ie, (T, t) no better than (S, s) for either player